

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0286

Shalersville 100 kV
August 19, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0286. The Interconnection Customer proposes a 113.4 MW battery energy storage facility interconnecting at the Shalersville 100 kV substation in the Ohio area. The request was received on November 13, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 163 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,944,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Chamberlin-Twinsburg 100kV Line 1
Contingency: Chamberlin-Twinsburg 100kV
Loading: 112.1% (Rating: 159 MVA)
DFAX: 10.1%
Pre-project loading: 105.0%
2. Facility: Twinsburg-Chamberlin 100kV
Contingency: Chamberlin-Twinsburg 100kV #1
Loading: 112.1% (Rating: 159 MVA)
DFAX: 10.1%
Pre-project loading: 105.0%
3. Facility: Tallmadge-David Swinford 100kV
Contingency: Weedpatch-Surrowiec 500kV
Loading: 103.9% (Rating: 158 MVA)
DFAX: 7.2%
Pre-project loading: 98.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chamberlin-Twinsburg 100kV Line 1	Rebuild line with higher-capacity conductor	\$2,938,000	100% (\$2,938,000)
Chamberlin-Tinsburg 100kV Cir. 2	Rebuild line with higher-capacity conductor	\$2,803,000	100% (\$2,803,000)
Tallmadge-David Swinford 100kV	Reconductor line	\$321,000	63% (\$203,000)
	Total allocated to project		\$5,944,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.